CancerOS — Synthetic Longitudinal Data Specification

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Subsystem: Synthetic Longitudinal Research Engine (SLRE)
Clinical scope: Breast cancer
Primary objective: Generate reproducible hypothetical patient trajectories from real clinical baselines

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1. Purpose and Design Philosophy

The Synthetic Longitudinal Research Engine (SLRE) shall generate time-indexed, clinically constrained research simulations from authentic breast cancer baseline records.

The system shall support alternative hypothetical tumor and patient-condition trajectories across different recorded or simulated treatment exposures.

The principal purpose is to demonstrate, test, and investigate the behavior of CancerOS as a longitudinal digital twin.

The simulator must not be presented as a validated patient-specific treatment response predictor.

1.1 Required capabilities

SLRE shall:

1. Accept a valid real breast cancer baseline.
2. Preserve its authentic source observations.
3. Evaluate the baseline for scenario eligibility.
4. Generate multiple hypothetical follow-up trajectories.
5. Represent different tumor response patterns.
6. Model treatment exposure as time-dependent information.
7. Simulate laboratory, vital-sign, and symptom changes.
8. Generate corresponding tumor geometry where supported.

9. Preserve uncertainty and missing information.
10. Record all simulation assumptions and parameters.
11. Validate generated observations before publication.
12. Support reproduction of previous simulation runs.

1.2 Scientific boundary

A synthetic trajectory describes a hypothetical scenario under specified modeling assumptions.

It does not prove:

- That a treatment caused a response.
- That the real patient would experience that outcome.
- That a simulated laboratory abnormality is a genuine adverse event.
- That a tumor's hypothetical growth establishes drug resistance.
- That a model trained on simulated histories will generalize to real patients.

The engine shall explicitly disclose these limitations.

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2. Simulation Architecture

SLRE shall use six logical processing layers.

Layer 1 — Real baseline adapter

Loads and validates:

- Source patient identity.
- Authentic imaging.
- Tumor segmentation or measurements.
- Available pathology information.
- Recorded clinical observations.
- Source provenance.

The real baseline is immutable.

Layer 2 — Clinical eligibility engine

Determines which hypothetical treatment-exposure configurations are compatible with known baseline information.

The engine shall reject incompatible configurations or mark them explicitly as unconstrained research hypotheticals that are excluded from clinically plausible scenario comparisons.

Layer 3 — Latent trajectory engine

Generates internal simulation variables describing:

- Tumor burden.
- Pattern of tumor change.
- Exposure timing.
- Treatment-related adverse-effect state.
- Functional patient-condition state.
- Model uncertainty.

These variables are simulation constructs, not direct observations of tumor biology.

Layer 4 — Observation generator

Produces dated:

- Tumor measurements.
- Laboratory values.
- Vital signs.
- Symptoms.
- Treatment records.
- Structured research notes.

Observations shall reflect sampling schedules, reporting variability, and explicitly modeled missingness.

Layer 5 — Imaging geometry generator

Generates synthetic follow-up lesion geometry from the original tumor segmentation.

It shall preserve traceability to the authentic baseline.

The resulting geometry must be labeled synthetic and must not be described as a real follow-up MRI.

Layer 6 — Validation and publication

Performs structural, temporal, numerical, clinical-consistency, and provenance validation.

Only successfully validated branches shall be available in the research viewer.

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3. Simulation Cohort Design

3.1 Initial targets

- Up to 100 authentic baseline patients.
- Five primary scenario families.
- One representative branch from each family per eligible baseline.
- Four observation states per branch, including baseline.
- Up to 500 synthetic branches.
- Up to 2,000 scenario-linked visit states.

These are capacity and demonstration targets.

A smaller validated cohort may be used during initial development.

3.2 Patient independence

All synthetic branches created from the same original patient shall retain a common parent source identity.

They shall not count as independent real patients.

A baseline patient and every descendant branch must remain in the same training, validation, or test partition.

3.3 Branch identity

Every branch shall have:

- branch_id
- source_patient_id
- source_baseline_study_id

- scenario_family
- treatment_configuration_id
- simulation_version
- random_seed
- simulation_parameters
- generation_timestamp
- validation_status

A branch shall have an independent synthetic observation history.

The source baseline and other branches must remain unchanged.

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4. The Five Primary Clinical Scenario Families

The simulator shall implement five mandatory families.

These define hypothetical patterns rather than official clinical response classifications.

SC-01 — Strong tumor response

Objective

Demonstrate sustained reduction in the tumor measurement during the simulated treatment period.

Required behavior

- Tumor volume or selected tumor-burden measure declines.
- Follow-up changes remain ordered according to the scenario definition.
- Clinical symptoms may improve, remain stable, or fluctuate.
- Treatment-related laboratory abnormalities may occur independently of tumor shrinkage.

Example demonstration trajectory

Baseline volume: 12.4 cm^3
Day 30: 9.2 cm^3

Day 60: 5.7 cm^3
Day 90: 3.1 cm^3

These numbers are illustrative, not medically validated treatment-response parameters.

Required interpretation

The AI may state that the **simulated lesion measurement decreases over the represented period**.

It shall not declare a formal complete or partial response without an appropriate response-assessment method and supporting criteria.

SC-02 — Partial improvement or relative stability

Objective

Demonstrate limited reduction or minimal change in tumor measurements.

Required behavior

- Tumor measurements change modestly.
- Small variations may fall within modeled measurement uncertainty.
- Patient-condition observations evolve independently.
- The simulator records whether the intended trajectory is modest improvement or near stability.

Example trajectory

Baseline: 12.4 cm^3
Day 30: 11.5 cm^3
Day 60: 10.3 cm^3
Day 90: 9.6 cm^3

Required interpretation

The AI shall describe the measured or simulated changes rather than automatically assign a formal clinical response category.

SC-03 — Progressive tumor burden

Objective

Demonstrate an increasing lesion measurement despite a hypothetical recorded treatment exposure.

Required behavior

- Tumor burden rises across selected follow-up observations.
- Symptoms may worsen, fluctuate, or remain unchanged.
- The simulated treatment history remains visible.
- The engine shall not automatically infer treatment resistance or causation.

Example trajectory

Baseline: 12.4 cm^3
Day 30: 14.1 cm^3
Day 60: 16.8 cm^3
Day 90: 19.5 cm^3

Required interpretation

The AI may flag the increasing simulated lesion measurement for research review.

It must not assert an established biological mechanism of progression.

SC-04 — Initial response followed by regrowth

Objective

Demonstrate a change in tumor trajectory direction.

Required behavior

- Initial observations show reduction.
- A later observation shows renewed increase.
- The inflection in the synthetic trajectory is preserved.
- Each observation is available for comparison.
- The treatment-event history remains visible.

Example trajectory

Baseline: 12.4 cm^3
Day 30: 8.6 cm^3
Day 60: 7.9 cm^3
Day 90: 11.4 cm^3

Required interpretation

The AI may identify regrowth following an earlier simulated reduction.

It shall not claim confirmed acquired drug resistance.

SC-05 — Clinical deterioration or treatment-related toxicity scenario

Objective

Demonstrate that the patient's clinical condition may worsen independently of tumor burden.

Required behavior

- Tumor measurements may shrink, remain stable, or increase.
- Selected laboratory values may become abnormal.
- Symptoms may worsen.
- Vital signs may demonstrate plausible changes.
- Adverse events shall use documented terminology where implemented.
- The system shall separate observed deterioration from hypothetical causal attribution.

Example trajectory

Tumor volume:

- Day 0: 12.4 cm^3
- Day 30: 12.2 cm^3
- Day 60: 12.1 cm^3
- Day 90: 12.3 cm^3

Additional synthetic observations may include fatigue, changing blood counts, or other configured adverse-event features.

Required interpretation

The AI may explain that tumor burden appears relatively stable in the synthetic scenario while selected clinical indicators worsen.

The system must not equate stable tumor burden with overall patient stability.

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5. Independent Tumor and Toxicity Dimensions

The five scenario families are narrative templates.

Underlying tumor behavior and clinical deterioration shall be modeled as separate dimensions.

5.1 Tumor dimension

Allowed categories:

- Sustained shrinkage.
- Limited shrinkage.
- Relative stability.
- Continued growth.
- Shrinkage followed by regrowth.

5.2 Clinical-condition dimension

Allowed categories:

- Relatively stable clinical condition.
- Mild deterioration.
- Moderate deterioration.
- Severe simulated deterioration.
- Improvement following a previously simulated adverse effect.

Severity labels are internal simulation categories unless explicitly mapped to a supported clinical grading standard.

5.3 Combined states

Examples:

- Tumor shrinking with stable blood counts.
- Tumor shrinking with worsening blood counts.
- Tumor stable with severe fatigue.
- Tumor growing with relatively stable vital signs.
- Tumor growing with clinical deterioration.
- Regrowth after earlier improvement with or without an adverse-event history.

This representation prevents inappropriate assumptions that tumor shrinkage always means improved overall health.

6. Treatment Exposure Modeling

Treatment shall be modeled as a time-indexed exposure and intervention history.

A treatment label does not automatically determine its effectiveness.

6.1 TE-01 — Chemotherapy

Supported synthetic features:

- Start and stop events.
- Regimen category.
- Treatment cycles when explicitly configured.
- Treatment interruptions.
- Tumor-burden trajectories.
- Selected laboratory and symptom scenarios.

The system shall not invent patient-specific dosage recommendations.

6.2 TE-02 — Endocrine therapy

The compatibility engine shall require documented compatible hormone-receptor status for clinically constrained endocrine-therapy scenarios.

Unknown receptor status shall not be silently assumed positive.

6.3 TE-03 — HER2-targeted therapy

The compatibility engine shall require an appropriate HER2 status and treatment context.

The simulation may represent tumor changes and separately configured adverse-effect scenarios.

It shall not use HER2-targeted therapy indiscriminately across all source cases.

6.4 TE-04 — Immunotherapy-containing treatment

The simulator shall require an explicitly defined compatible breast cancer subtype and research treatment setting.

It shall not assume immunotherapy is suitable for all breast cancer diagnoses.

6.5 TE-05 — Surgery and/or radiotherapy

Surgical intervention shall be modeled as a discrete event.

Surgical resection shall not be represented merely by continuously shrinking an unresected tumor mesh.

Postoperative representations shall distinguish residual disease measurements from removal of the original lesion.

Radiotherapy may be represented as a dated exposure category where scientifically and contextually appropriate.

6.6 Combined and sequential therapies

The engine may support:

- Sequential treatment categories.
- Combination regimens.
- Treatment interruptions.
- Changes between treatment phases.

Each event shall carry the documented or hypothetical status and source provenance.

6.7 Treatment eligibility matrix

The simulator shall maintain machine-readable rules for:

- Receptor status requirements.
- Cancer subtype compatibility.
- Disease setting.
- Treatment intent.
- Treatment sequencing assumptions.
- Relevant contraindication or exclusion conditions explicitly supported by the configured knowledge base.
- Missing required clinical information.

These rules require review against authoritative clinical sources before being considered clinically grounded.

The hackathon system shall not imply clinical appropriateness solely because a scenario passes an internal software rule.

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7. Simulation Time Model

7.1 Default demonstration observations

Default relative observation days:

- Day 0.
- Day 30.
- Day 60.
- Day 90.

These dates are interface demonstration defaults, not recommended clinical follow-up schedules.

7.2 Irregular observation support

The data structure shall support nonuniform time intervals.

For example:

- Day 0.
- Day 21.
- Day 52.
- Day 91.

The simulator must not assume that all patients are tested at fixed intervals.

7.3 Modality-specific observation schedules

MRI, laboratory tests, symptoms, and treatment events may occur on different dates.

CancerOS shall maintain each event's own timestamp.

When data are synchronized into a visit summary, the system must retain the original dates.

7.4 Missing observations

The simulator may intentionally omit selected measurements.

Missing values must not be replaced with normal results.

Missingness shall be represented through explicit masks or absent observations, accompanied by a configured reason where known.

8. Tumor Trajectory Simulation

8.1 Tumor state

For the MVP, the primary simulated anatomical variable may be tumor volume.

Additional features may include:

- Longest diameter.
- Orthogonal dimensions.
- Derived geometric shape.
- Tumor position relative to the source imaging coordinate system.
- Measurement uncertainty.

Volume and diameter must remain distinct quantities.

8.2 Baseline initialization

The simulator shall initialize from a verified lesion mask or measurement.

Where only a bounding box exists, the engine shall not label a derived approximate volume as accurately segmented tumor volume.

8.3 Trajectory function

The engineering model may define a latent state:

$V(t) = \max(0, V0 \times \exp(\int r(s)\, ds))$

Where:

- V0 is the baseline tumor-volume parameter.
- r(s) is a configurable hypothetical net-change function.
- t is simulation-relative time.

This mathematical expression is a proposed simulation mechanism.

It is not an established biological model for an individual breast cancer patient.

The engine may use simpler deterministic interpolating functions for the curated demonstration.

8.4 Scenario-specific trajectory constraints

Strong response: Net reduction over the designated simulation horizon.

Partial response or stable: Restricted or modest movement according to the chosen template.

Progression: Increasing lesion burden over the designated interval.

Initial response then regrowth: At least one decline followed by an increase.

Clinical deterioration: Tumor trajectory specified independently of clinical-condition trajectory.

8.5 Random variability

The simulator shall optionally incorporate seeded stochastic variation.

Variability may affect:

- Visit measurements.
- Observation timing.
- Laboratory values.
- Symptom intensity.
- Missingness.
- Measurement noise.

Generated noise shall not violate nonnegativity, valid units, or explicitly defined constraints.

Noise distributions shall be documented and tested.

8.6 Uncertainty

Three uncertainty sources shall be represented separately where supported:

- Measurement uncertainty.
- Simulation-parameter uncertainty.
- Predictive uncertainty.

A deterministic scenario trajectory must not automatically be assigned a clinically meaningful confidence interval.

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9. Laboratory Data Simulation

9.1 Supported variables

Initial laboratory variables:

- Hemoglobin.
- White blood cell count.
- Absolute neutrophil count.
- Platelets.
- Creatinine.
- Selected liver-function measurements.

Each simulated variable shall use a validated dictionary entry containing:

- Analyte code.
- Canonical unit.
- Allowed numeric representation.
- Context-specific reference interval source.
- Missingness handling.
- Applicable scenario constraints.

9.2 Temporal behavior

Laboratory simulations shall allow:

- Relative stability.
- Gradual changes.
- Transient deviations.
- Recovery patterns.
- Persistent abnormalities.
- Measurement variability.

The generator shall not make every laboratory parameter follow tumor growth monotonically.

9.3 Correlation rules

Related variables may share latent influences where justified.

Examples of proposed model relationships:

- Total WBC and absolute neutrophil count are related measurements.
- Platelet and hemoglobin changes may coexist with a treatment-related marrow-suppression scenario.
- Clinical deterioration may occur without parallel tumor growth.

The engine must use explicit correlation configurations rather than unconstrained independent random values.

9.4 Temporal timing

Laboratory changes shall be associated with the relevant hypothetical event schedule.

Treatment exposure shall not automatically cause an abnormal result.

9.5 Laboratory validity

Validation shall reject:

- Wrong units.
- Unparseable values.
- Invalid temporal ordering.
- Nonfinite numerical values.
- Impossible negative counts.
- Absent analyte definitions.
- Invalidly mixed values from different simulation branches.

Out-of-reference-range values are not automatically invalid.

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10. Vital-Sign and Symptom Simulation

10.1 Vital signs

Supported parameters:

- Heart rate.
- Blood pressure.
- Temperature.
- Weight.

Each variable shall use configurable, source-supported constraints.

The generator may simulate deterioration patterns, but shall not independently produce a clinical diagnosis from an abnormal vital sign.

10.2 Symptoms

Initial symptoms:

- Fatigue.
- Pain.
- Nausea.
- Swelling.

Additional symptoms may be configured as structured entities.

10.3 Severity representation

Every symptom shall specify the scale being used.

The engine shall not assume that a generic 0–10 symptom rating is equivalent to an official oncology adverse-event grade.

10.4 Temporal symptom course

Symptoms may:

- Appear.
- Increase.
- Decrease.
- Resolve.
- Recur.
- Remain missing when not recorded.

10.5 Consistency

The engine shall avoid logical contradictions, such as a symptom being simultaneously marked absent and severe within the same timestamped observation without a documented reconciliation.

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11. Treatment Toxicity and Adverse Events

11.1 Terminology standard

The simulator should use the National Cancer Institute's Common Terminology Criteria for Adverse Events (CTCAE) version 6.0 where appropriate.

CTCAE v6.0 was released in 2025.

11.2 Important constraint

The presence of a laboratory abnormality is not always sufficient to assign an adverse-event grade.

The exact adverse-event definition, applicable baseline measurements, clinical context, and version-specific grading rules must be considered.

11.3 Required fields

A simulated adverse-event record shall contain:

- Event type.
- Onset date.
- Resolution date, if applicable.
- Grade or ungraded status.
- Evidence inputs.
- Treatment-exposure context.
- Simulation provenance.
- Ruleset version.

11.4 Safety boundaries

A simulated event shall never be described as an actual adverse drug reaction in the original patient.

Causal attribution shall remain hypothetical unless the record represents a genuine documented source finding.

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12. Synthetic Imaging and Geometry

12.1 Purpose

The geometry engine shall create a visual representation of hypothetical tumor changes from a real baseline segmentation.

It does not need to produce photorealistic follow-up MRI images for the MVP.

12.2 Baseline geometry

Start from:

- Authentic breast MRI geometry.
- Verified or documented tumor segmentation.
- Baseline physical spacing and orientation.
- Lesion identity.

12.3 Follow-up representation

The engine may generate:

- Expanded lesion mask.
- Contracted lesion mask.
- Constrained geometric shape changes.
- Synthetic 3D lesion mesh.
- Linked tumor measurements.

The generator shall preserve a traceable mapping to the source coordinate system.

12.4 Geometric consistency

Required validation:

- Nonempty lesion masks where a lesion is represented.
- Valid 3D coordinates.
- Consistent image spacing.
- Correct lesion volume calculations.
- No NaN or infinite coordinates.
- No unexplained geometric jumps between states.

- Anatomical boundary checks where the required tissue segmentation exists.
- Registration and transformation metadata retained.

12.5 Mesh generation

The implementation may use:

- Binary or labeled voxel masks.
- Surface extraction.
- Mesh smoothing with documented volume effects.
- Web-compatible mesh serialization.

The mesh shall retain links to the underlying synthetic mask and source image.

12.6 Unsupported biological visualization

The simulator shall not create purported factual maps of:

- Hypoxia.
- Molecular resistance.
- Drug penetration.
- Immune infiltration.
- Necrosis.
- Metastatic dissemination.

Any exploratory visualization of such properties would require a separate scientific model and explicit labeling.

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13. Synthetic Clinical Notes

13.1 Purpose

Generate short, structured narrative observations corresponding to the simulated measurements.

13.2 Generation method

The preferred approach shall be:

1. Generate structured simulation data.

2. Calculate numerical changes deterministically.
3. Assemble verified context.
4. Use controlled templates or an LLM to produce a readable research note.
5. Validate numerical consistency against structured inputs.

13.3 Prohibited note content

Notes shall not invent:

- A genuine physician encounter.
- An actual clinical test.
- A confirmed pathology finding.
- A treatment prescription.
- An unrecorded procedure.
- A verified diagnosis or adverse drug reaction.

Synthetic narrative notes shall identify themselves as simulated research observations.

13.4 Note format

A generated note may contain:

- Simulated observation date.
- Relevant tumor measurements.
- Laboratory observations.
- Symptoms.
- Treatment exposure context.
- Change from previous observation.
- Missing information.
- Research interpretation limitations.

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14. Formal Clinical Response Criteria

The simulator's five scenario families must not be confused with formal response categories.

14.1 RECIST 1.1

RECIST 1.1 uses standardized lesion selection and unidimensional measurements, with specific definitions for response and progression.

A percentage change in three-dimensional tumor volume is not automatically a RECIST response measurement.

14.2 MVP requirement

CancerOS shall use descriptive internal labels for the five synthetic scenario families.

It may calculate volumetric change as a research measurement.

Formal RECIST classifications shall only be implemented where the required measurements and assessment conditions are available.

14.3 Pathologic complete response

Pathologic complete response is a distinct histopathologic endpoint.

CancerOS shall not infer it solely from an MRI-derived synthetic tumor-volume curve.

14.4 Output language

Preferred:

"Simulated tumor volume decreased by 25% between the selected observations."

Not permitted without proper evidence:

"The patient achieved a confirmed RECIST partial response."

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15. Simulation Data Model

15.1 Simulation configuration

Required fields:

- simulation_config_id
- source_patient_id
- baseline_study_id
- scenario_family
- treatment_configuration

- observation_schedule
- tumor_model_parameters
- laboratory_model_parameters
- toxicity_configuration
- missingness_configuration
- noise_configuration
- seed
- simulator_version

15.2 Simulation run

Required fields:

- simulation_run_id
- simulation_config_id
- started_at
- completed_at
- status
- configuration_hash
- runtime_metadata
- validation_summary
- failure_reason

15.3 Generated visit

Required fields:

- synthetic_visit_id
- branch_id
- relative_day
- synthetic_date
- tumor_state_reference

- laboratory_record_references
- vital_record_references
- symptom_record_references
- treatment_event_references
- provenance_type

All generated observations shall carry SYNTHETIC_GENERATED provenance unless they are explicitly derived measurements, interpolation outputs, or model predictions with their own corresponding classifications.

15.4 Event model

Events shall be stored individually rather than encoded only inside a single visit snapshot.

Supported event categories:

- Imaging observation.
- Laboratory observation.
- Vital-sign measurement.
- Symptom report.
- Treatment event.
- Adverse-event observation.
- Clinical narrative observation.
- Research outcome observation.

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16. Simulation Algorithm Requirements

The implementation shall follow the sequence below.

Algorithm SLRE-GEN-001

Input: Approved baseline patient, scenario configuration, random seed.

Procedure:

1. Retrieve the approved source patient.
2. Verify access and provenance.

3. Load baseline tumor measurements and optional segmentation.
4. Load known clinical eligibility variables.
5. Validate the proposed scenario configuration.
6. Initialize the deterministic random-number generator.
7. Construct the hypothetical treatment-event timeline.
8. Generate the latent tumor trajectory.
9. Generate the independent clinical-condition and toxicity trajectory.
10. Generate modality-specific observation schedules.
11. Sample supported laboratory, vital-sign, and symptom observations.
12. Produce synthetic lesion geometry where required.
13. Generate structured research notes from validated simulated values.
14. Calculate derived measurements.
15. Attach provenance and configuration metadata.
16. Execute validation rules.
17. Publish the branch if all blocking checks pass.
18. Otherwise quarantine the branch and report errors.

Output: Validated synthetic longitudinal branch or structured generation failure.

16.1 Transactional behavior

Incomplete branches shall not become visible as completed simulations.

A failed generation run must not modify the authentic baseline.

16.2 Reproducibility

The same source data, versioned configuration, seed, numerical environment, and simulator version should reproduce the same outputs within documented tolerances.

16.3 Simulator versions

Any change in clinical rules, parameter distributions, geometry processing, or simulation mathematics shall create a new version.

Existing published branches shall retain their original version references.

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17. Validation Framework

Validation shall operate at multiple levels.

Level V1 — Structural validation

Checks:

- Required fields.
- Schema correctness.
- Unique identifiers.
- Valid references.
- Recognized categorical values.

Level V2 — Temporal validation

Checks:

- Valid timestamps.
- Chronological event ordering.
- Treatment timing.
- Observation schedules.
- Branch date relationships.

Level V3 — Numerical validation

Checks:

- Valid units.
- Finite numbers.
- Nonnegative quantities where required.
- Valid geometry and physical spacing.
- Plausible bounds according to documented variable constraints.

Level V4 — Scenario validation

Checks:

- Intended tumor trajectory family.
- Independent clinical-condition state.
- Treatment configuration consistency.
- Required inflection points where applicable.
- Absence of contradictory scenario labels.

Level V5 — Cross-modal validation

Checks:

- Imaging-derived volume matches generated mask within tolerance.
- Tumor measurements align with the selected visit.
- Clinical observations belong to the correct branch.
- Laboratory observations use valid measurement definitions.
- Clinical narrative numbers match structured records.
- Treatment events align with configured exposure windows.

Level V6 — Provenance validation

Checks:

- Authentic observations retain their true source.
- Every generated observation identifies its simulator.
- Parent patient and branch references are present.
- Source license and transformation history remain linked.
- No synthetic result is marked as authentic.

Level V7 — Scientific and safety validation

Checks:

- No unsupported treatment efficacy claims.
- No unsupported molecular resistance claims.
- No formal response classification without required assessment information.
- No fabricated real patient outcomes.

- No unsupported causal attribution for adverse effects.

Level V8 — Reproducibility validation

Checks:

- Version and seed stored.
- Configuration hash stored.
- Repeated run produces matching intended outputs.
- Generation failures can be reconstructed.

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18. Automated Test Cases

The minimum simulation test suite shall include:

Test ID	Scenario	Expected behavior
SLRE-T01	Valid strong-response configuration	Generates declining target tumor trajectory
SLRE-T02	Valid stable-disease configuration	Generates constrained near-stable measurements
SLRE-T03	Valid progression configuration	Generates increasing target tumor trajectory
SLRE-T04	Initial response then regrowth	Preserves reduction and later increase
SLRE-T05	Toxicity with stable tumor	Generates deterioration independently of tumor burden
SLRE-T06	Toxicity with tumor shrinkage	Both outcomes represented simultaneously
SLRE-T07	Missing receptor status	Rejects or flags treatment configurations requiring known status
SLRE-T08	Incompatible HER2 configuration	Blocks clinically constrained generation

SLRE-T09	Invalid laboratory unit	Validation failure
SLRE-T10	Negative tumor volume	Validation failure
SLRE-T11	Out-of-order observation dates	Validation failure
SLRE-T12	Missing optional blood result	Preserves missingness
SLRE-T13	Identical seed and configuration	Reproducible outputs
SLRE-T14	Different branch of same patient	Baseline unchanged
SLRE-T15	Different patient identity	No record crossover
SLRE-T16	Failed generation	No partially published branch
SLRE-T17	Incorrect synthetic provenance	Blocking validation failure
SLRE-T18	Narrative conflicts with structured record	Rejects or regenerates narrative
SLRE-T19	Invalid lesion mask geometry	Blocks affected imaging output
SLRE-T20	Formal RECIST label without required evidence	Label withheld
SLRE-T21	Surgery event	Uses a discrete intervention transition
SLRE-T22	Irregular observation dates	Correctly preserves actual intervals

SLRE-T23	Unavailable AI service	Structured simulated records remain accessible
SLRE-T24	Source dataset removed or restricted	Derived branch follows access and retention policy

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19. Evaluation Methodology

19.1 Technical correctness evaluation

Measure:

- Successful generation rate.
- Validation pass rate.
- Schema integrity.
- Reproducibility.
- Time-order correctness.
- Unit correctness.
- Provenance completeness.
- Geometry consistency.
- Cross-modal numerical agreement.

19.2 Synthetic distribution evaluation

Where sufficient real reference data exist, compare selected synthetic statistics with actual observations.

Possible comparisons:

- Measurement distributions.
- Temporal autocorrelation.
- Changes between observations.
- Correlation among laboratory measurements.
- Missingness characteristics.
- Variability of imaging measurements.

The comparison must be limited to variables genuinely present in the reference data.

19.3 Clinical plausibility review

A clinical reviewer should inspect representative scenarios where feasible.

Review topics:

- Treatment-context compatibility.
- Implausible laboratory combinations.
- Inconsistent timelines.
- Symptom patterns.
- Unsupported response labels.
- Misleading medical interpretations.

No claim of medical realism shall be based only on the absence of software validation errors.

19.4 Real-data evaluation

The genuine longitudinal evaluation cohort shall remain separate from synthetic trajectory evaluation.

Use authentic follow-up imaging to assess selected capabilities, such as:

- Segmentation agreement.
- Volume measurement error.
- Tumor change detection.
- Limited temporal estimation error where justified.

Synthetic labels cannot establish real-world treatment-response prediction accuracy.

19.5 Baselines

Where forecasting is attempted, compare against simple methods such as:

- Last-observation-carried-forward.
- Linear extrapolation.
- A clearly specified simple temporal regression.

A complex temporal model shall not be presented as superior without corresponding held-out evidence.

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20. Performance and Resource Constraints

The simulator shall be designed for relatively modest hardware.

20.1 CPU execution

Tabular simulation and core validation shall run on CPU without requiring a dedicated GPU.

20.2 Imaging efficiency

Large MRI volumes shall be processed separately from routine clinical record generation.

Curated lesion meshes and preview artifacts may be precomputed.

20.3 Batch processing

The engine shall support generation of one branch at a time and small batches.

20.4 Memory limits

The system shall avoid loading the entire imaging cohort into memory.

Images and masks shall be loaded lazily or processed in bounded batches.

20.5 Service isolation

A simulation failure shall not prevent access to existing patient records and previously generated branches.

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21. User Interface Requirements

The simulation workspace shall provide:

1. Baseline patient selection.
2. Eligible scenario-family selection.
3. Treatment-exposure configuration.
4. Observation schedule.
5. Simulation status.
6. Validation results.
7. Generated branch identifier.

8. Access to the timeline and 3D viewer.
9. Explicit synthetic-data labels.
10. Simulation assumptions and uncertainty.
11. Branch-to-branch comparison where implemented.

The interface shall not imply that selecting a different treatment route generates a clinically validated estimate of its effectiveness for the source patient.

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22. Deliverables

SLRE implementation shall produce:

- Simulation configuration schema.
- Five scenario-family definitions.
- Treatment compatibility rules.
- Tumor trajectory generator.
- Laboratory observation generator.
- Vital-sign generator.
- Symptom generator.
- Adverse-event representation.
- Treatment-event generator.
- Synthetic lesion geometry generator.
- Structured narrative generator.
- Provenance and versioning module.
- Validation engine.
- Batch generation interface.
- Synthetic branch database tables.
- Automated testing suite.
- Curated simulation demonstration cases.

- Technical and scientific limitations report.

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23. Acceptance Criteria

SLRE will be considered hackathon-ready when all of the following are met.

AC-S01 — Five scenario families

Every required family can generate a complete curated scenario.

AC-S02 — Authentic baseline preservation

No simulation changes the original patient measurements or source imaging.

AC-S03 — Longitudinal completeness

Each curated demonstration branch contains at least four dated observation states, including its baseline reference.

AC-S04 — Cross-modal integrity

Tumor measurements, associated geometry, laboratories, symptoms, and treatment events are correctly associated with their intended observation times.

AC-S05 — Treatment eligibility

Required treatment-compatibility checks are enforced.

AC-S06 — Provenance completeness

All synthetic observations identify their source branch and generation configuration.

AC-S07 — Reproducibility

Repeating the same approved configuration reproduces the same intended records within documented tolerances.

AC-S08 — Numerical integrity

All curated generated records pass blocking numerical and schema checks.

AC-S09 — Scientific interpretation boundaries

The generated records and AI interpretations do not misrepresent hypothetical scenarios as validated treatment effects.

AC-S10 — Digital twin compatibility

Generated lesion states and clinical observations can be loaded through the ordinary CancerOS timeline and visualization interfaces.

AC-S11 — Branch comparison

At least two distinct hypothetical branches can be inspected without cross-contamination or modification of the authentic source baseline.

AC-S12 — Evaluation separation

Synthetic simulator testing and real patient longitudinal evaluation are reported independently.

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24. Scientific References and Implementation Standards

The following resources govern terminology or provide relevant external evidence.

RECIST 1.1 — Response Evaluation Criteria in Solid Tumours

https://recist.eortc.org/recist-1-1/

Used for distinguishing formal clinical response criteria from CancerOS's descriptive synthetic tumor-volume scenarios.

NCI Common Terminology Criteria for Adverse Events (CTCAE), Version 6.0

https://dctd.cancer.gov/research/ctep-trials/for-sites/adverse-events

Used for standardized adverse-event terminology and grading where the required evidence is available.

FDA — Addressing the Limitations of Medical Data in AI

https://www.fda.gov/medical-devices/medical-device-regulatory-science-research-programs-conducted-osel/addressing-limitations-medical-data-ai

Provides relevant background on the opportunities and limitations of supplementing medical AI datasets with synthetic information.

Real longitudinal imaging references

I-SPY2 and I-SPY1 from TCIA, as defined in COS-SPEC-003.

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25. Final Engineering Directive

The Synthetic Longitudinal Research Engine must be implemented as a separate, versioned research subsystem.

It shall generate structured observations before generating explanatory text or visual representations.

The system shall:

1. Start from verified source data.
2. Apply explicit compatibility constraints.
3. Generate hypothetical time-dependent states.
4. Produce measurable clinical observations.
5. Generate corresponding synthetic lesion geometry.
6. Record the full provenance chain.
7. Validate results.
8. Publish only acceptable research branches.
9. Expose assumptions and uncertainty.
10. Remain scientifically distinguishable from observed patient outcomes.

The goal is not to imitate a real patient history so convincingly that its artificial origin becomes invisible.

The goal is to create a controlled, transparent, reproducible environment in which CancerOS can demonstrate and evaluate longitudinal cancer-data processing.

End of COS-SPEC-004 — Version 1.0